Summary

Immunology deals with a study of the immune system. The immune system recognises and eliminates the invaders, and the ability of the body to overcome the pathogen is called immunity. Immunity is classified into innate immunity and acquired immunity. Acquired immunity is further classified into cell mediated immunity and antibody mediated immunity as its components. Acquired immunity may be active or passive immunity. Immune response is the body's response to pathogens and it may be primary or secondary. The organs involved in the origin, maturation and proliferation of lymphocytes are called lymphoid organs. Thymus, bone marrow are primary lymphoid organs. The secondary lymphoid organs are lymph node, MALT, GALT and BALT.

An antigen is a substance that is specific to an antibody. An immunogen is a substance capable of initiating an immune response. Haptens are substance that are non-immunogenic but can react with the product of a specific immune response. Substances that can enhance the immune response to an antigen are called adjuvants. An epitope is also known as antigenic determinant and is the active part of the antigen. A paratope is the part of antibody. Precipitation, agglutination, neutralization, opsonisation etc., are the different types of antigen and antibody reaction. A vaccine is biological preparation that provides active acquired immunity. The malfunctioning of immune system leads to hypersensitivity, immunodeficiency or autoimmune diseases. A tumour or neoplasm is a group of cells whose growth has gone unchecked. The best results in the treatment of cancer is achieved by an integrated approach to therapy, surgery, radiotherapy, chemotherapy and immunotherapy.

Evaluation

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1. Colostrum provides
a) Naturally acquired active immunity
b) **Naturally acquired passive immunity**
c) Artificially acquired active immunity
d) Artificially acquired passive immunity

2. Paratope is an
(A) Antibody binding site on variable regions
(B) Antibody binding site on heavy regions
(C) Antigen binding site on variable regions
(D) Antigen binding site on heavy regions

3. Allergy involves
a) IgE b) IgG
c) lgA d) IgM

4. Anaphylactic shock is due to
(A) Allergic reaction
(B) Secretion of toxins
(C) Secretion of histamines
(D) All the above

5. Spread of cancerous cells to distant sites is termed as
(A) Metastasis
(B) Oncogenes
(C) Proto-oncogenes
(D) Malignant neoplasm

6. AIDS virus has
(A) Single stranded RNA
(B) Double stranded RNA
(C) Single stranded DNA
(D) Double stranded DNA

7. All are peripheral lymphoid organs except
(A) Lymph nodes
(B) Spleen
(C) Mucosa associated lymphoid tissue
(D) Thymus

8. Which is not a macrophage?
(A) Monocyte (B) Microglia
(C) Kupffer cell **(D) Lymphocyte**

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